A Quick Look at Data Mining with Weka

With an abundance of data from different sources, data mining for various purposes is the rage these days. Weka is a collection of machine learning algorithms that can be used for data mining tasks. It is open source software and can be used via a GUI, Java API and command line interfaces, which makes it very versatile.

Waikato Environment for Knowledge Analysis (Weka) is free software licensed under the GNU General Public License. It has been developed by the Department of Computer Science, University of Waikato, New Zealand. Weka has a collection of machine learning algorithms including data preprocessing tools, classification/regression algorithms, clustering algorithms, algorithms for finding association rules, and algorithms for feature selection. It is written in Java and runs on almost any platform.

Let's look at the various options of machine learning and data mining available in Weka and discover how the Weka GUI can be used by a newbie to learn various data mining techniques. Weka can be used in three different ways – via the GUI, a Java API and a command line interface. The GUI has three components—Explorer, Experimenter and Knowledge Flow, apart from a simple command line interface.

The components of Explorer

Explorer has the following components.

Preprocess: The first component of Explorer provides an option for data preprocessing. Various formats of data like ARFF, CSV, C4.5, binary, etc, can be imported. ARFF stands for attribute-relation file format, and it was developed for use with the Weka machine learning software. Figure 1 explains various components of the ARFF format. This is an example of the Iris data set which comes along with Weka. The first part is the relation name. The 'attribute' section contains the names of the attributes and their data types, as well as all the actual instances. Data can also be imported from a URL or from a SQL database (using JDBC). The Explorer component provides an option to edit the data set, if required. Weka has specific tools for data preprocessing, called filters.

The filter has two properties: supervised or unsupervised. Each supervised and unsupervised filter has two categories, attribute filters and instances filters. These filters are used to remove certain attributes or instances that meet a certain condition. They can be used for discretisation, normalisation, resampling, attribute selection, transforming and combining attributes. Data discretisation is a data reduction technique, which is used to convert a large domain of numerical values to categorical values.